

CHEMBL230 Raw Dataset Analysis

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Yes. I have now **actually read and analyzed the uploaded** CHEMBL230_RAW_DATA.csv, not just the parent project's description.

The most important thing for you to understand before explaining it to **Swathi mam and Srikanth sir** is this:

This is a raw bioactivity dataset. It is not yet the final machine-learning dataset.

Our first job is to understand what is inside it, identify missing/inconsistent records, and decide how to preprocess it before calculating molecular descriptors and training XGBoost.

1. What is this dataset?

The file is:

CHEMBL230_RAW_DATA.csv

It contains **7,979 bioactivity records** and **5 columns**.

The five columns are:

Column	Meaning
Molecule ChEMBL ID	Unique identifier assigned to a molecule in ChEMBL
Smiles	Chemical structure represented as a SMILES string
Standard Type	Type of biological activity measurement
Standard Value	Numerical value of that activity measurement
Standard Units	Unit in which the activity is reported

The dataset is specifically associated with the **COX-2 target / CHEMBL230** project that we're building.

2. The most important concept: what is a bioactivity record?

Think of one row as:

"For this particular molecule, an experiment measured its activity against the target and obtained this value."

For example, the first row in your actual dataset is:

Molecule ChEMBL ID : CHEMBL70104

SMILES:
CC1(C)OC(=O)C(OC2CCCCC2)=C1c1ccc(S(C)(=O)=O)cc1

Standard Type : IC50

Standard Value : 40

Standard Units : nM

So we can interpret this as:

CHEMBL70104 is the molecule, its chemical structure is represented by the SMILES, and its reported **IC50 is 40 nM**.

That is a very important record for our project because IC50 is one of the measurements we can use to characterize biological activity.

3. Dataset dimensions

The first thing we should always check in an unfamiliar dataset is:

```
df.shape
```

Your result is:

```
(7979, 5)
```

Meaning:

- **7,979 rows**

- **5 columns**

So there are 7,979 records in the raw file.

Important clarification

7,979 rows does **not** mean 7,979 unique molecules.

This is one of the most important observations in your dataset.

4. Number of unique molecules

We found:

```
Unique Molecule ChEMBL IDs = 6,073
```

while there are:

```
Total records = 7,979
```

So some molecules appear multiple times.

In fact:

```
1,003 molecule IDs occur more than once.
```

The most repeated molecule is:

```
CHEMBL118 → 218 records
```

Other examples include:

```
CHEMBL6 → 121 records
```

```
CHEMBL122 → 63 records
```

```
CHEMBL521 → 37 records
```

```
CHEMBL139 → 26 records
```

Why?

This is **not automatically an error**.

A molecule can have multiple experimental bioactivity records.

For example:

```
Molecule A
```

```
↓
```

```
Experiment 1 → IC50 = 40 nM
```

```
Experiment 2 → IC50 = 50 nM
```

```
Experiment 3 → IC50 = 60 nM
```

Therefore:

One molecule can have multiple measurements.

This is why we should not simply say "duplicate molecule = delete it."

We need to decide how to aggregate or filter repeated measurements during preprocessing.

5. SMILES — one of the most important columns

The Smiles column represents the molecular structure.

For example:

```
CC1(C)OC(=O)C(OC2CCCCC2)=C1c1ccc(S(C)(=O)=O)cc1
```

This isn't meant to be read like normal English.

SMILES stands for:

Simplified Molecular Input Line Entry System

It provides a machine-readable representation of a chemical structure.

Why do we need it?

Because later we use:

SMILES → RDKit → molecular descriptors

For example:

```
SMILES
  ↓
RDKit
  ↓
Molecular Weight
LogP
TPSA
H-bond donors
H-bond acceptors
Rotatable bonds
  ↓
Machine Learning
  ↓
XGBoost
```

This is one of the central ideas of your entire project.

6. SMILES quality in your dataset

Your dataset has:

```
7,789 non-missing SMILES
190 missing SMILES
```

Percentage missing:

```
≈ 2.38%
```

I also checked the **7,789 available SMILES using RDKit**.

Result:

```
Valid SMILES: 7,789
Invalid SMILES: 0
```

That's a useful observation.

So:

Among the records where a SMILES is present, the SMILES strings are RDKit-valid.

But the **190 missing SMILES** are still a problem because we cannot calculate molecular descriptors for those records.

7. Standard Type

The Standard Type column contains:

```
IC50
```

for **7,822 records**.

There are:

```
157 missing values
```

Importantly, there isn't a mixture of:

IC50
EC50
Ki
Kd

among the non-missing records.

Instead, the available activity type is overwhelmingly:

IC50

This is useful because we're not dealing with multiple biological measurement types that need to be mixed together.

8. What exactly is IC50?

This is something you should be able to explain to your mentor.

IC50 = Half Maximal Inhibitory Concentration

It represents the concentration of a compound required to inhibit a biological activity by approximately **50%**, under the conditions of the particular assay.

In general:

Lower IC50
↓
Less compound concentration needed
↓
Stronger inhibition

and:

Higher IC50
↓
More compound concentration needed
↓
Weaker inhibition

Simple example

Suppose:

Compound A → IC50 = 20 nM
Compound B → IC50 = 5000 nM

Under comparable assay conditions, Compound A shows stronger inhibition than Compound B.

Important: We should not blindly compare every measurement across different experimental conditions, which is one reason preprocessing matters.

9. Standard Units — very important problem

Your dataset contains these units:

nM
ug.mL-1
ug
%

Counts:

Unit	Records
nM	6,832
ug.mL-1	25
ug	1
%	1
Missing	1,120

This is an important preprocessing issue.

Why?

Suppose we have:

10 nM

and:

10 ug.mL⁻¹

The numerical value 10 is not directly comparable because the units represent different quantities.

Therefore:

We cannot treat all Standard Value numbers as if they are directly comparable.

For our eventual IC50-based classification, we need to carefully decide what records to retain and whether/how to convert units.

10. Why nM is particularly useful here

The overwhelming majority of measured records with units are:

nM

Specifically:

6,832 records

This gives us a relatively consistent group for the main activity analysis.

The parent project also uses a threshold based on **10,000 nM** to define active/inactive compounds.

But that classification should happen **after appropriate preprocessing**, not on the raw file blindly.

11. Standard Value

Standard Value is the numerical activity measurement.

In your dataset:

Non-missing Standard Value = 6,840
Missing Standard Value = 1,139

So approximately:

14.27% missing

This is significant.

We cannot calculate activity-based labels for records where the activity value is missing.

12. Range of activity values

For all non-missing Standard Value records:

Minimum = 0
Maximum ≈ 833,335,724.3

The mean is approximately:

1,063,770

while the median is:

970.5

This huge difference is extremely important.

Why?

Because the distribution is **strongly right-skewed**.

There are some extremely large activity values.

For example:

```
Median ≈ 970.5
Mean ≈ 1,063,770
```

If the distribution were relatively balanced, mean and median would usually be much closer.

Here they are extremely different because of large values/outliers.

13. Why raw IC50 values can be difficult for ML

Suppose the dataset contains:

```
10
20
50
100
1000
10000
100000
5000000
```

The values span several orders of magnitude.

This creates a highly skewed distribution.

In pharmacological datasets, a common approach is to work with a logarithmic activity scale such as:

```
pIC50 = -log10(IC50 in molar units)
```

But **we should not automatically apply this yet**.

First we need to establish exactly what preprocessing methodology your project is going to follow.

For your current task, the important thing to tell your mentor is:

"The raw Standard Value distribution is highly skewed, so we need appropriate preprocessing before modeling."

14. Zero values

I found:

```
6 records
```

where:

```
Standard Value = 0
```

This needs attention.

A zero IC50 value is problematic for transformations such as:

```
log10(value)
```

because:

```
log10(0)
```

is undefined.

Therefore these records need to be investigated during preprocessing rather than silently transformed.

15. Negative values

I checked for negative activity values.

Result:

```
Negative Standard Values = 0
```

That's good.

16. Missing data — complete picture

Here is the actual missing-data situation:

Column	Missing	Percentage
Molecule ChEMBL ID	0	0%
Smiles	190	2.38%
Standard Type	157	1.97%
Standard Value	1,139	14.27%
Standard Units	1,120	14.04%

This gives us an immediate preprocessing plan.

The most problematic field is:

Standard Value

because almost **14.3%** of records don't have an activity value.

17. There are also completely duplicated rows

I found:

602 exact duplicate rows

That means 602 rows are repeated exactly across all five columns.

The dataset contains:

7,979 total rows

and:

7,377 unique rows

after removing exact duplicate rows.

Important distinction

There are two different concepts:

Exact duplicate

Everything is identical:

ID
SMILES
Type
Value
Units

→ likely safe candidate for duplicate removal.

Repeated molecule

Same molecule but potentially different experimental measurements:

CHEMBL118

appearing many times.

→ **do not automatically delete these.**

This distinction is important when explaining preprocessing to your mentor.

18. Repeated SMILES

There are:

```
5,927 unique SMILES
```

among the dataset.

There are:

```
2,051 repeated SMILES records
```

Again, this doesn't necessarily mean bad data.

The same chemical structure may have multiple experimental measurements.

19. Interesting observation about molecule IDs and SMILES

I checked whether a molecule ID maps to multiple different SMILES.

Result:

```
IDs with multiple different non-missing SMILES = 0
```

So within this dataset, the same ChEMBL ID does not appear to be associated with different non-missing structures.

That's useful for data consistency.

20. Missing-data patterns

There are **931 records** where:

Smiles	present
Standard Type	present
Standard Value	missing
Standard Units	missing

So these are essentially:

```
Molecule
  ↓
Structure available
  ↓
IC50 type available
  ↓
Actual numerical activity unavailable
```

These records may be useful for some structure-level analysis, but **they cannot directly contribute to an IC50-based supervised classification label** unless we have another legitimate source for the activity.

21. Activity threshold used in the parent project

The parent methodology we previously examined uses:

```
IC50 < 10,000 nM → Active
IC50 ≥ 10,000 nM → Inactive
```

This is important, but here's the distinction you should make to your mentors:

The raw dataset does not itself contain an Active/Inactive column.

We create that target during preprocessing using the chosen threshold.

For the nM records with non-missing values, the threshold produces:

```
IC50 < 10,000 nM → 4,978 records
IC50 ≥ 10,000 nM → 1,835 records
```

Total:

```
6,813 records
```

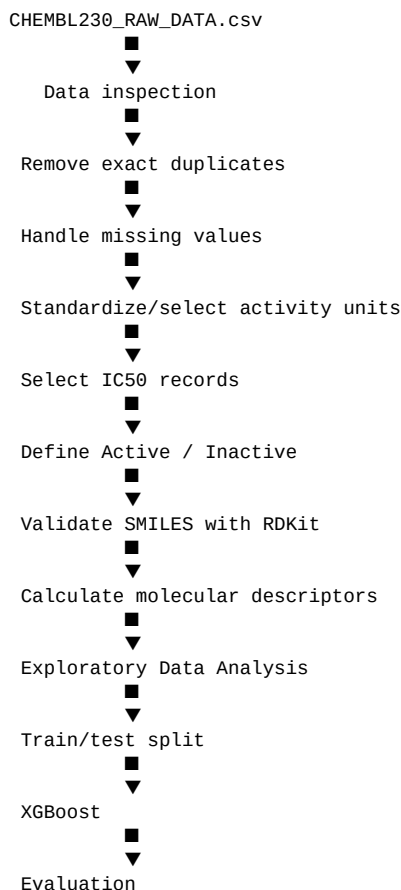
So approximately:

73.1% below 10,000 nM
26.9% at/above 10,000 nM

But don't call these final model classes yet. This is the result of applying the parent project's threshold to the currently usable nM measurements; duplicate handling, molecule-level aggregation, and other preprocessing decisions still matter.

22. What happens to the raw dataset?

Our conceptual pipeline is:



This is the actual logic you should understand.

23. What are molecular descriptors?

After we clean the dataset, we take the SMILES and use **RDKit** to calculate numerical properties.

For the parent project, the six important descriptors are:

1. Molecular Weight

Approximate mass of the molecule.

SMILES → RDKit → Molecular Weight

2. Rotatable Bonds

Measures flexibility of the molecule based on rotatable single bonds.

More rotatable bonds generally indicate greater structural flexibility.

3. Hydrogen Bond Acceptors

Number of atoms/sites capable of accepting a hydrogen bond.

4. Hydrogen Bond Donors

Number of atoms/sites capable of donating a hydrogen bond.

5. TPSA

Topological Polar Surface Area

It provides an estimate of the polar surface characteristics of a molecule.

6. LogP

A measure related to the compound's preference for:

lipid-like environment ↔ aqueous environment

It is commonly used as an indicator of lipophilicity.

24. Why do we need descriptors?

XGBoost cannot directly understand:

CC1(C)OC(=O)...

as a chemical structure in the same way a chemist does.

We convert:

```
Chemical structure
  ↓
SMILES
  ↓
RDKit
  ↓
Numerical descriptors
  ↓
Machine-learning features
```

For example:

Molecule	MolWt	LogP	TPSA	HBD	HBA	Rotatable
A	350	2.8	55	1	4	5
B	420	4.1	72	2	6	7

Now XGBoost can work with the numerical representation.

25. Why XGBoost?

Our target eventually becomes:

Active
Inactive

This is a **classification problem**.

XGBoost is a gradient-boosted decision-tree algorithm.

Conceptually:

```
Descriptor data
  ↓
```

Decision tree
↓
Another tree learns previous errors
↓
Another tree
↓
Many trees combined
↓
Final prediction

For example:

Molecular Weight = 350
LogP = 3.1
TPSA = 60
HBD = 1
HBA = 4
Rotatable Bonds = 5
↓
XGBoost
↓
Active / Inactive

26. The most important things you can tell Swathi mam and Srikanth sir

If they ask:

"What did you find when you explored the dataset?"

You can say:

"I analyzed the raw CHEMBL230_RAW_DATA.csv. It contains 7,979 records and 5 columns: Molecule ChEMBL ID, SMILES, Standard Type, Standard Value and Standard Units. There are 6,073 unique molecule IDs, so multiple experimental records exist for some molecules. The available activity type is predominantly IC50, with 7,822 records labeled IC50 and 157 missing. Most measured units are nM, with 6,832 nM records, but there are also 25 ug/mL, one ug and one percentage record, so unit consistency needs to be handled during preprocessing."

Then continue:

"I found 190 missing SMILES, 1,139 missing Standard Values and 1,120 missing Standard Units. There are also 602 exact duplicate rows. The SMILES that are present are RDKit-valid. The Standard Value distribution is highly right-skewed, ranging from 0 to about 8.33×10^8 , so we need to handle outliers and the activity scale carefully before modeling."

That's a strong, factual dataset-exploration answer.

27. If they ask: "Why can't you directly train XGBoost?"

Say:

"Because the raw data is not yet model-ready. It contains missing values, mixed units, exact duplicates, repeated measurements for the same molecule, and raw chemical structures in SMILES format. We first need to clean and standardize the bioactivity data, define the target classes, validate the structures, calculate molecular descriptors using RDKit, and then use those numerical descriptors as input features for XGBoost."

That is the key answer.

28. If they ask: "What is the biggest issue you found?"

Don't say simply "missing data."

A better answer is:

"The main issue is that the raw dataset contains heterogeneous experimental records rather than one clean row per molecule."

Why?

Because we have:

```
7,979 records
↓
6,073 unique molecules
```

and some molecules have many measurements.

We therefore need to carefully decide how to handle:

```
duplicate measurements
missing activity
mixed units
multiple measurements per compound
```

before creating the final ML dataset.

29. What we should NOT do yet

Don't immediately do:

```
df.dropna()
```

and call the result clean.

Don't immediately do:

```
df.drop_duplicates()
```

and assume every repeated molecule is an error.

Don't immediately classify every raw number as:

```
<10000 = Active
>=10000 = Inactive
```

because the raw file contains non-nM units and missing values.

And don't calculate descriptors before deciding which records survive preprocessing.

30. Our next project step

Now that we've actually understood the raw dataset, the next logical stage is:

Dataset Exploration Notebook

We should create:

```
XGBioactive/
├──
│   ├── data/
│   │   └── CHEMBL230_RAW_DATA.csv
│   ├── notebooks/
│   │   └── 01_Dataset_Exploration.ipynb
│   ├──
│   ├── models/
│   ├── results/
│   ├── src/
│   ├── app/
│   ├── README.md
│   └── .gitignore
```

The notebook should document the findings we just made:

1. Import libraries
2. Load dataset
3. Display first 5 rows
4. Shape
5. Columns
6. Data types
7. Missing values
8. Unique molecules
9. Duplicate records
10. Standard Type distribution
11. Standard Units distribution
12. Standard Value statistics

13. Activity-value distribution
14. SMILES validation
15. Initial observations
16. Preprocessing decisions to be made

Do not modify the raw CSV. Keep CHEMBL230_RAW_DATA.csv as our original/raw source. Any cleaned dataset we create later should be a separate file.

And all of this work should go to your branch-karthikeya, while main remains untouched.

Today 1:53 PM